Ethan Hung

ethanhung720@gmail.com | github.com/ethanhung2007 | linkedin.com/in/ethanhung0720

EDUCATION

Georgia Institute of Technology *Atlanta, GA*
Bachelor of Science in Computer Engineering *August 2026 - Expected May 2028*

University of Illinois Urbana-Champaign *Urbana, IL*
Bachelor of Science in Computer Engineering *August 2025 - May 2026*

- GPA: 4.0/4.0

Relevant Coursework: Computer Systems and Programming, Digital System Design, Electrical & Electronic Circuits

EXPERIENCE

Design Verification Member *Atlanta, Georgia*
SiliconJackets *Sep 2026 - Present*

- Executed a full RTL to GDS flow for a pipelined RISC-V CPU in SystemVerilog on the SkyWater 130nm node
- Constructed a UVM-style testbench with constrained random tests and SVA, reaching code coverage of above 98%
- Inserted clock gating and high-speed cell variants, reducing negative slack by 28% compared to the baseline

Research and Development Intern *Taipei, Taiwan*
National Taiwan University *May 2026 - Aug. 2026*

- Developed a hardware-calibrated, trace driven SoC simulator modeling CIM, CPU, GPU, and NPU focused on optimizing LLM inference, achieving within 10% variance from Axellera Metis silicon
- Validated end-to-end simulator accuracy against Microsoft's Phi-3 Mini for prefill TTFT and decode throughput
- Profiled LLM operator streams and tensor shapes to model compute workloads, input activation, and memory transfer
- Verified Cortex A76 and Mali G610 latency via benchmarks on custom gem5 configurations and Mali Offline Compiler

Circuit Design Project Lead *Urbana-Champaign, IL*
Illini Electric Motorsports — Formula SAE *Sep. 2025 - May 2026*

- Designed a 4-layer vehicle control unit integrating sensor filtering, fault triggering, and vehicle control systems
- Engineered component placement and trace routing around vehicle subsystem locations and mechanical constraints
- Debugged VCU hardware failures, tracing PWM buzzer errors and inductor overheating due to component selection
- Implemented RC filters with op-amp buffers to condition brake pressure and current sense signals for MCU ADC inputs
- Coordinated circuit interfaces with STM32 I/O mapping with software for CAN, PWM, and vehicle control signals

PROJECTS

SIMT GPU Streaming Multiprocessor

- Architected a 4-warp × 4-lane SIMT GPU with round-robin scheduling, multi-lane execution, and per-warp control
- Defined a custom 32-bit ISA with predication, memory ops, branching, reconvergence, and MFSR instructions
- Built fetch/decode and scoreboard-based RAW hazard handling with warp stalls and control-flow recovery
- Incorporated predicate divergence and reconvergence, measuring 83.3–100% lane utilization across 3 GPU kernels
- Confirmed assembly-to-RTL execution on an Arty-A7-100T to 18.9K LUTs, 22.8K FFs, and 12 DSPs at 50 MHz

Processing in Memory FPGA Accelerator

- Designed an eight-tile split-K matrix multiplication accelerator in SystemVerilog on an Artix-7 operating at 158.7 MHz
- Implemented BRAM-local DSP compute units with row-stationary dataflow, reducing input-row memory traffic by 8×
- Increased arithmetic intensity from 0.50 to 0.89 ops/byte through local operand reuse across eight compute tiles
- Developed a pipelined reduction tree and double-buffered BRAM; increased throughput by 1.57× to 1.935 GOPS

PATENTS

Zero Weight Feeling Computer Mouse

- Conceptualized a floating mouse set combining air flow and magnetic repulsion to minimize friction
- Granted patent by the Taiwan Intellectual Property Office (Patent No. M663224)

SKILLS AND COMPETENCIES

Programming: SystemVerilog, Verilog, VHDL, C/C++, Embedded C, Python, RISC-V Assembly, TCL, Java, Makefiles
Skills: UVM, SystemVerilog Assertions (SVA), RTL Design & Validation, Static Timing Analysis (STA)
Software: Linux, Git, Cadence, Xilinx Vivado, Intel Quartus, Altium Designer, LTSpice, Librelane (Sky130 PDK)